

Detector Acceptance Study (Existing Vs Latest)

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- Re-Calculate Acceptance Correction for each roadset (long term)
- Currently calculated acceptance corrections by using Kenichi's new root files
- Compare Latest Acceptance Correction Vs Existing Acceptance Correction

Files Used & Kinematic Bins

Latest Files Used (Kenichi):

- **Thrown:** `rs67_<lh2/ld2>_acc.root:tree_4pi:event`
- **Accepted:** `rs67_<lh2/ld2>_acc.root:tree_acc:event`

Existing Files Used (Hugo):

- **Thrown:** `mc_drellyan_<LH2/LD2>_M027_S001_4pi_pTxFweight_v2.root`
- **Accepted:** `mc_drellyan_<LH2/LD2>_M027_S001_clean_occ_pTxFweight_v2.root`

Kinematic Bin Boundaries:

- **Mass [GeV]:** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.8]
- x_F : [0.0, 0.05, ..., 0.80]
- p_T [GeV/c]: [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]

Event Selection Criteria (Chuck's Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.05 < x_T \leq 0.58$
- $38 < dp_z < 116 \text{ GeV}$
- $dp_x^2 + dp_y^2 < 5.0 \text{ GeV}^2$

Target & Vertex:

- $-280 < dz < -5 \text{ cm}$
- $|dx| < 0.25 \text{ cm}$
- $|dy - 1.6| < 0.22 \text{ cm}$

Track Quality:

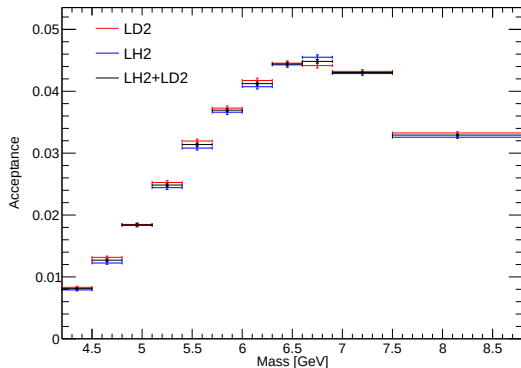
- $\chi_{\text{dimuon}}^2 < 18$
- Track separation $< 270 \text{ cm}$
- $p_{z,\text{st1}} \in [9, 75] \text{ GeV}$
- $n\text{Hits}_1 > 13, n\text{Hits}_2 > 13$

Generator-Level Fiducial Cuts (Thrown):

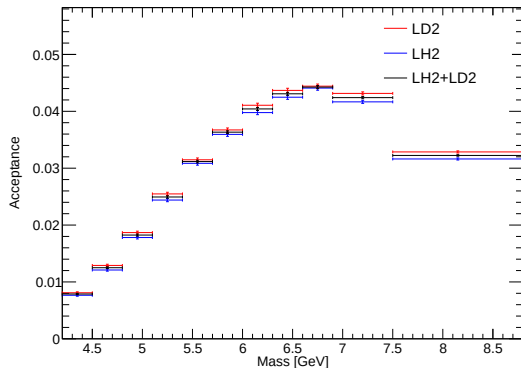
- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.0 < p_T \leq 3.0 \text{ GeV}/c$

Existing (Left) vs Latest (Right): Mass (Integrated)

Integrated Acceptance vs mass

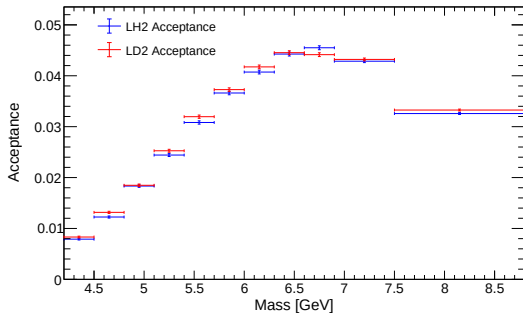


Integrated Acceptance vs mass

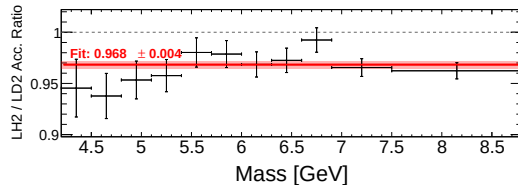
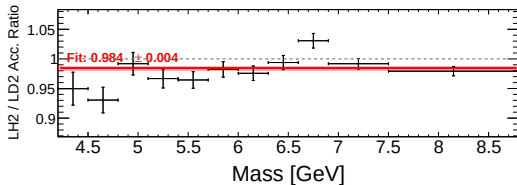
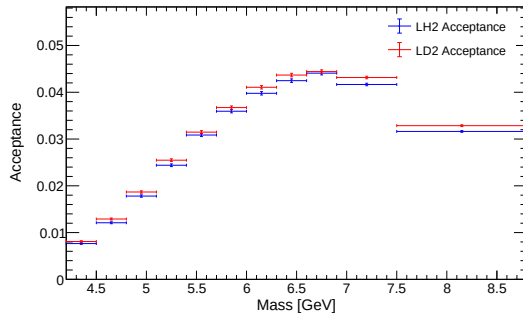


Existing (Left) vs Latest (Right): Mass Acceptance (All x_F , p_T)

Invariant Mass Acceptance (All x_F , p_T)

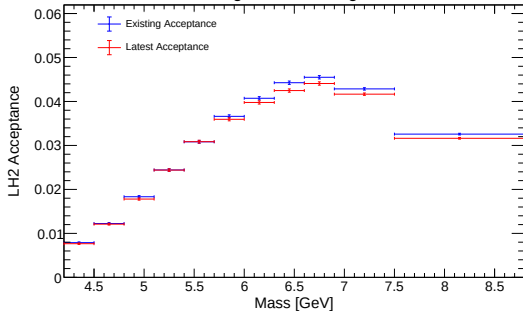


Invariant Mass Acceptance (All x_F , p_T)

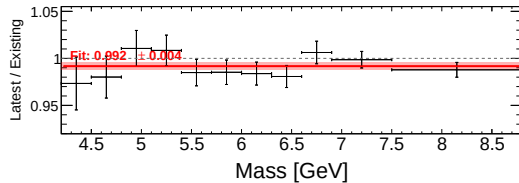
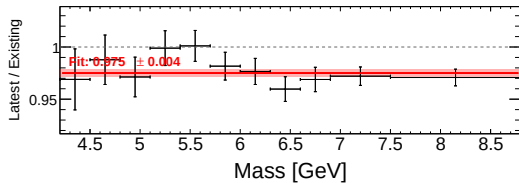
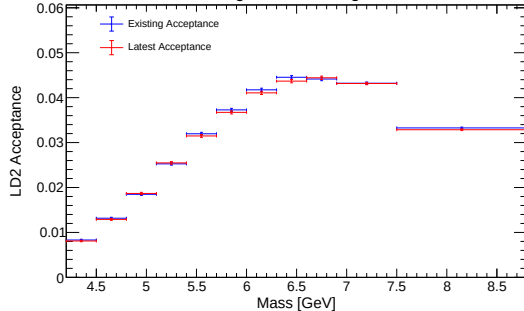


LH2 (Left) vs LD2 (Right): Latest/Existing Mass Acc.

LH2 Existing vs Latest: Integrated Mass

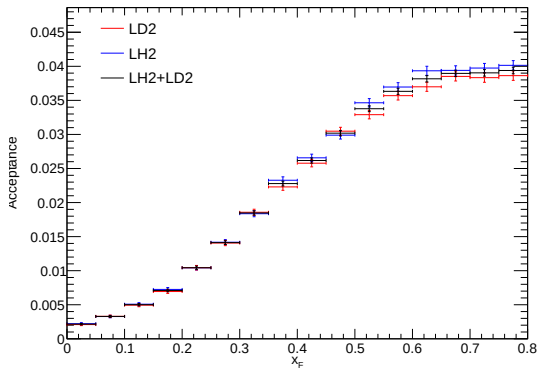


LD2 Existing vs Latest: Integrated Mass

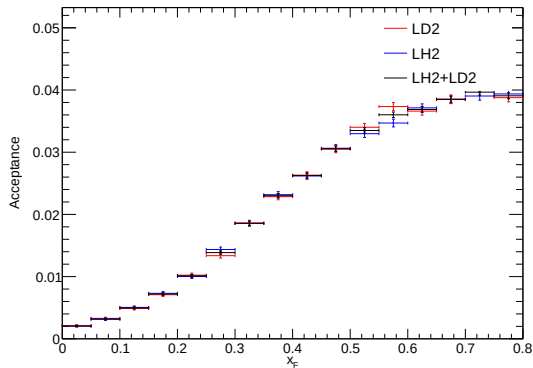


Existing (Left) vs Latest (Right): x_F (Integrated)

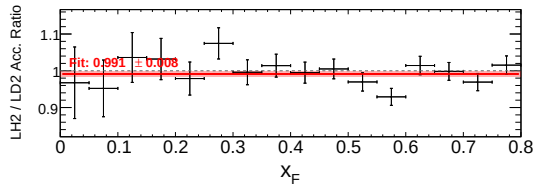
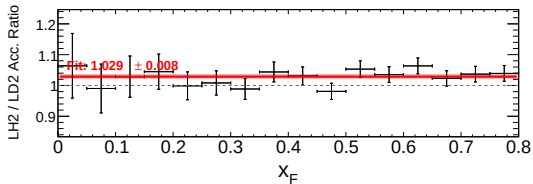
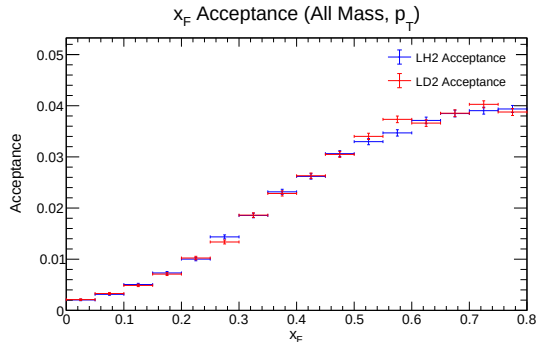
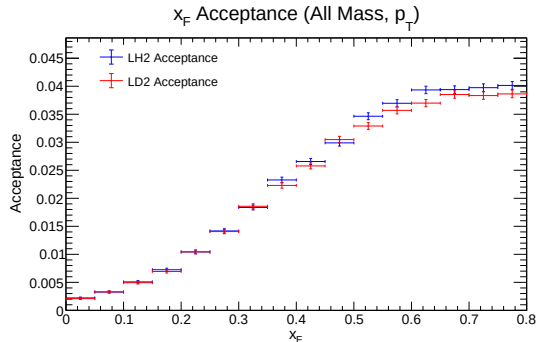
Integrated Acceptance vs x_F



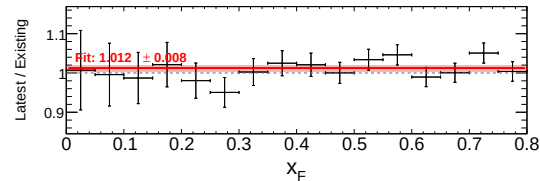
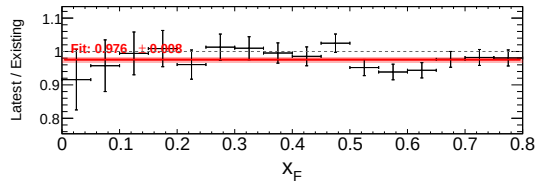
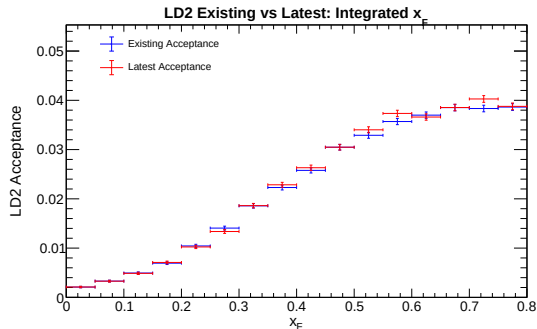
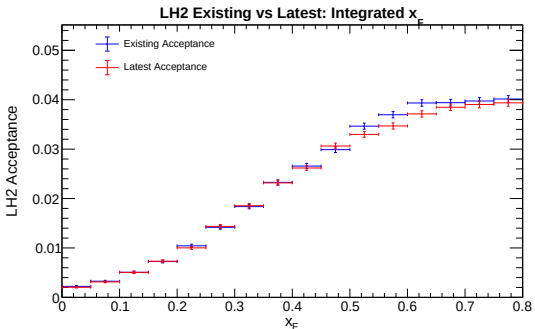
Integrated Acceptance vs x_F



Existing (Left) vs Latest (Right): x_F Acceptance (All Mass, p_T)

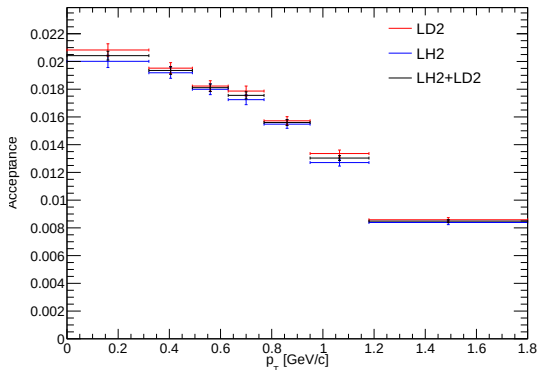


LH2 (Left) vs LD2 (Right): Latest/Existing x_F Acc.

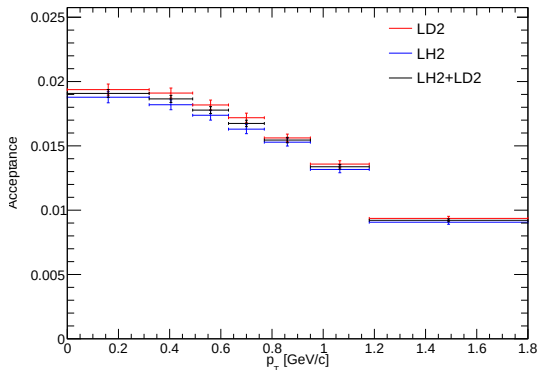


Existing (Left) vs Latest (Right): p_T (Integrated)

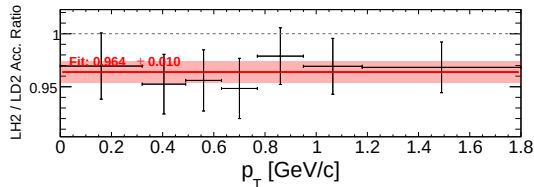
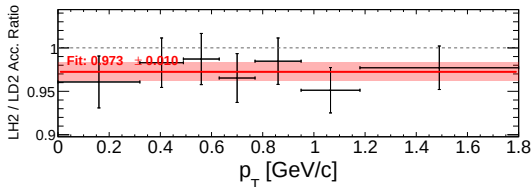
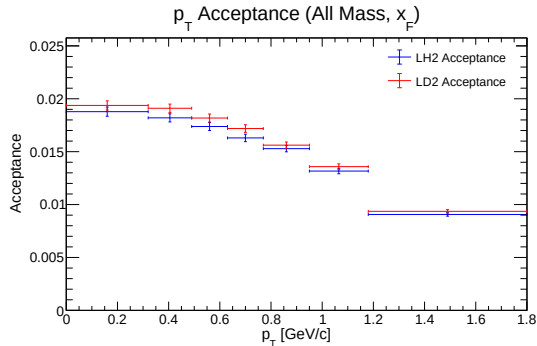
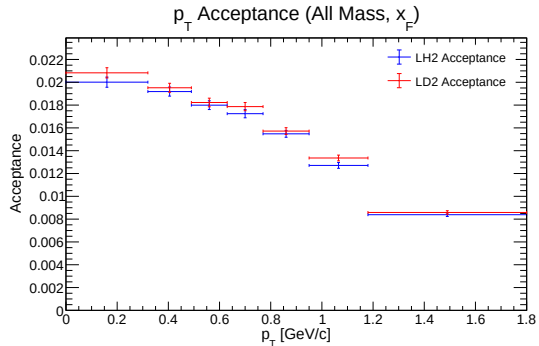
Integrated Acceptance vs p_T



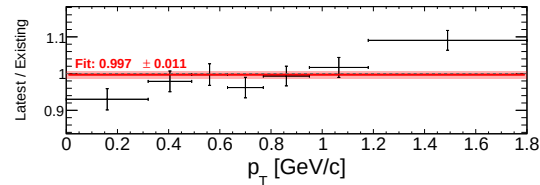
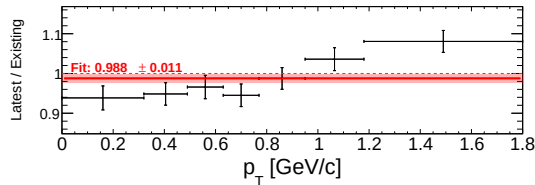
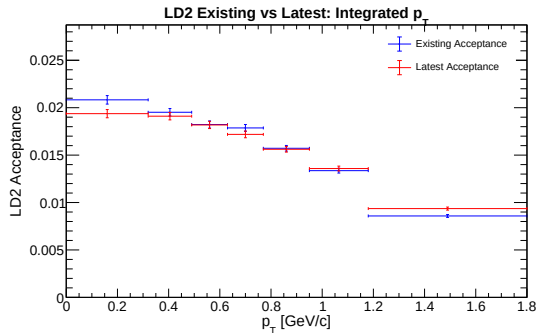
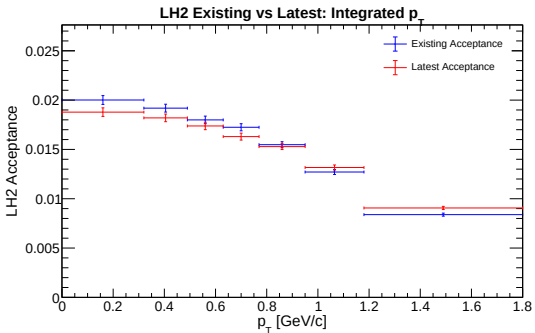
Integrated Acceptance vs p_T



Existing (Left) vs Latest (Right): p_T Acceptance (All Mass, x_F)

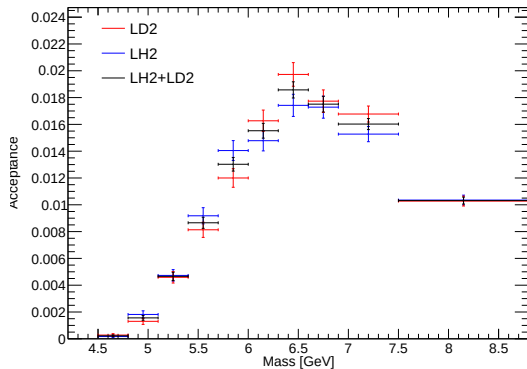


LH2 (Left) vs LD2 (Right): Latest/Existing p_T Acc.

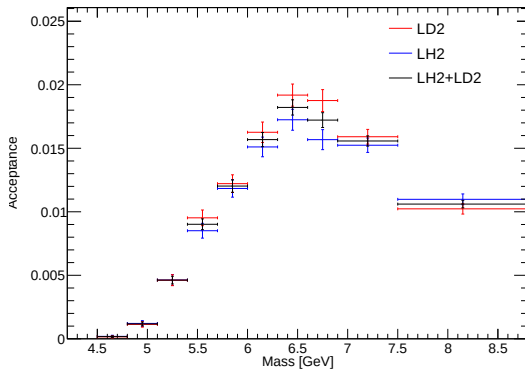


Existing vs Latest: Mass Overlay ($0.00 \leq x_F < 0.05$)

$0.00 \leq x_F < 0.05$

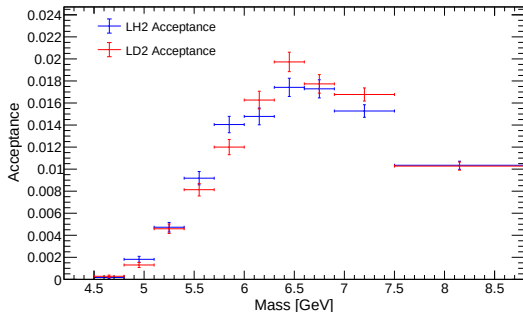


$0.00 \leq x_F < 0.05$

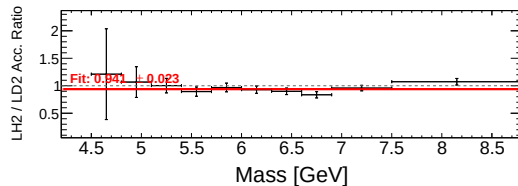
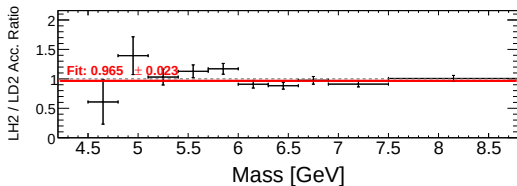
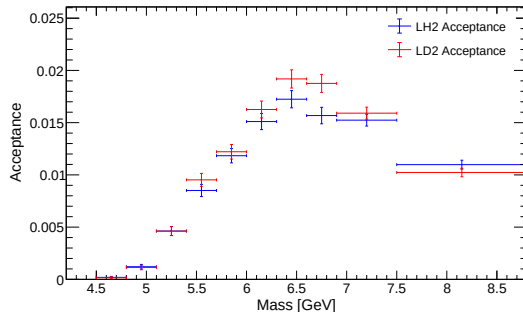


Existing vs Latest: Mass Acceptance ($0.00 \leq x_F < 0.05$)

Invariant Mass Acceptance ($0.00 \leq x_F < 0.05$)

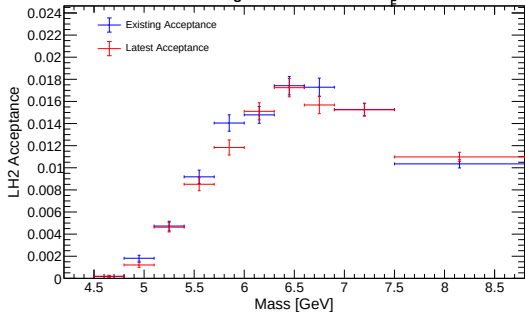


Invariant Mass Acceptance ($0.00 \leq x_F < 0.05$)

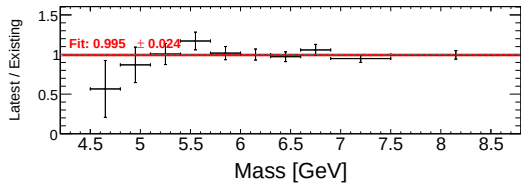
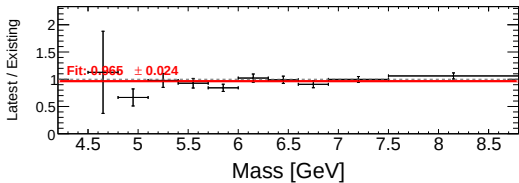
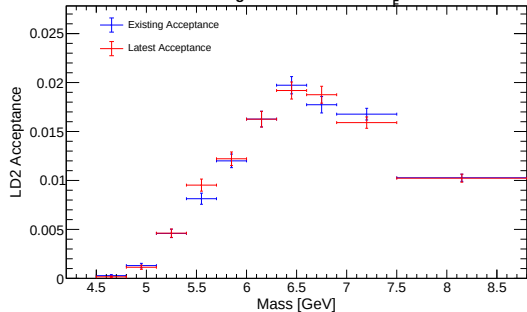


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.00 \leq x_F < 0.05$)

LH2 Existing vs Latest: Mass in x_F Bin 0

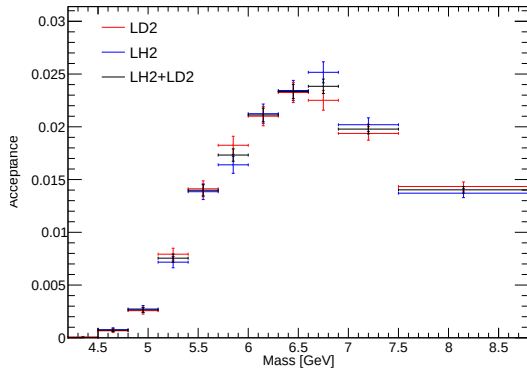


LD2 Existing vs Latest: Mass in x_F Bin 0

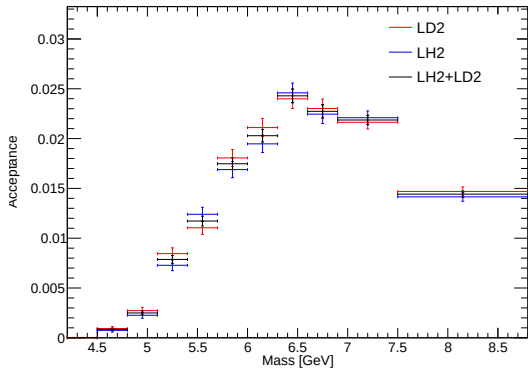


Existing vs Latest: Mass Overlay ($0.05 \leq x_F < 0.10$)

$0.05 \leq x_F < 0.10$

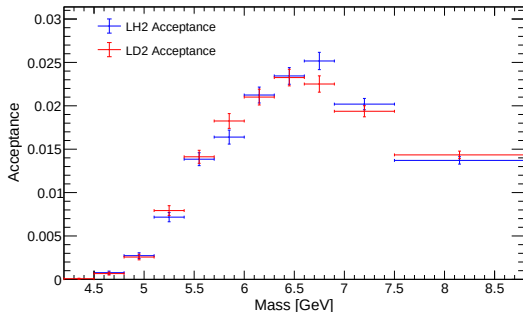


$0.05 \leq x_F < 0.10$

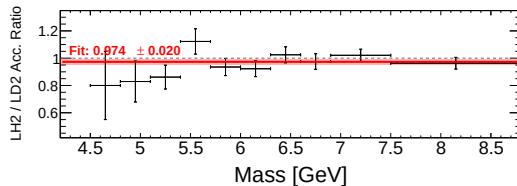
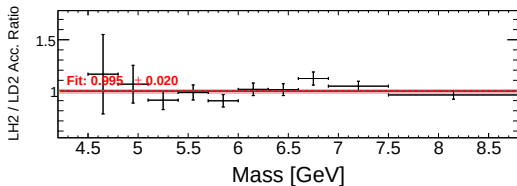
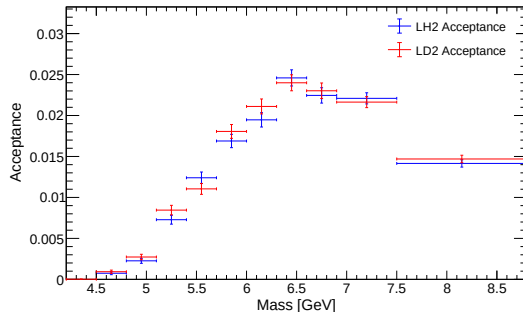


Existing vs Latest: Mass Acceptance ($0.05 \leq x_F < 0.10$)

Invariant Mass Acceptance ($0.05 \leq x_F < 0.10$)

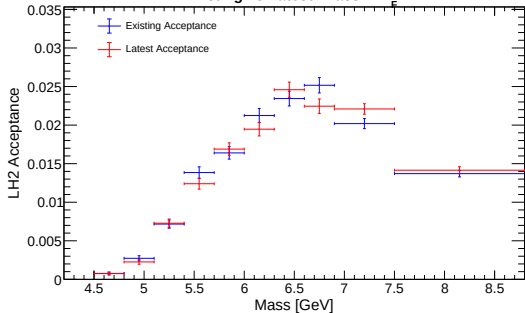


Invariant Mass Acceptance ($0.05 \leq x_F < 0.10$)

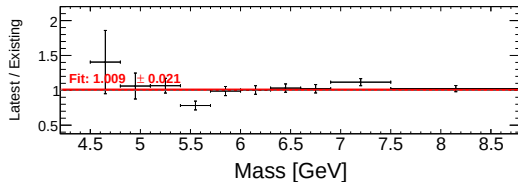
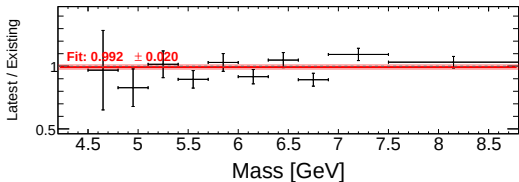
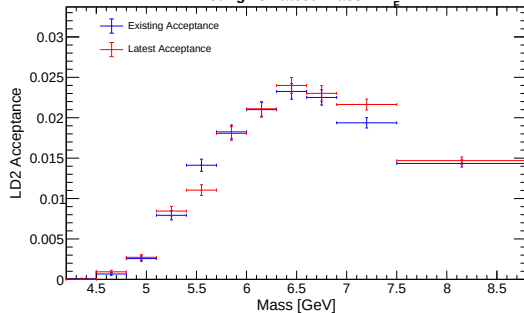


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.05 \leq x_F < 0.10$)

LH2 Existing vs Latest: Mass in x_F Bin 1

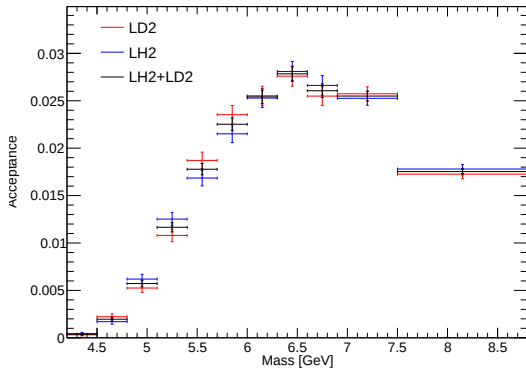


LD2 Existing vs Latest: Mass in x_F Bin 1

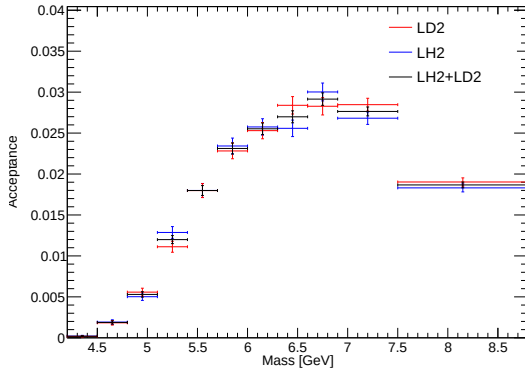


Existing vs Latest: Mass Overlay ($0.10 \leq x_F < 0.15$)

$0.10 \leq x_F < 0.15$

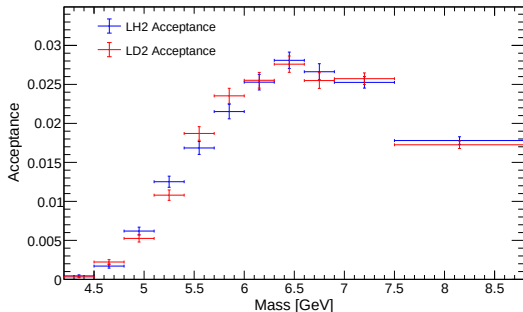


$0.10 \leq x_F < 0.15$

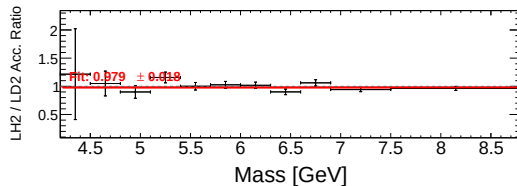
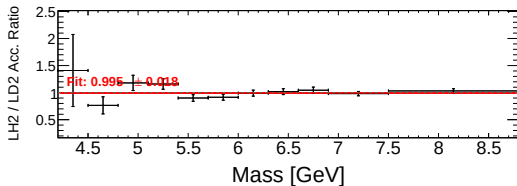
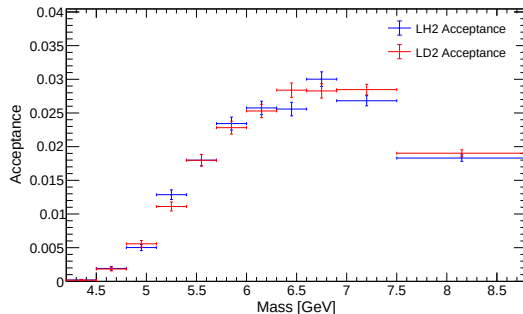


Existing vs Latest: Mass Acceptance ($0.10 \leq x_F < 0.15$)

Invariant Mass Acceptance ($0.10 \leq x_F < 0.15$)

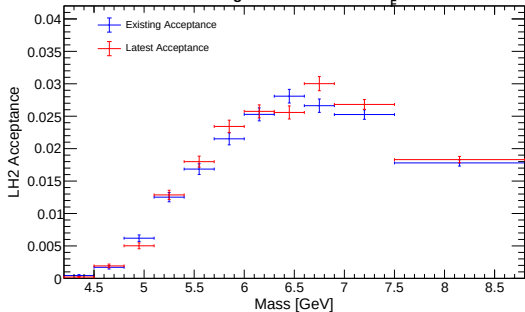


Invariant Mass Acceptance ($0.10 \leq x_F < 0.15$)

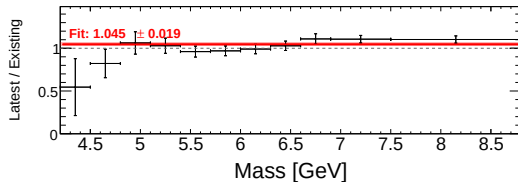
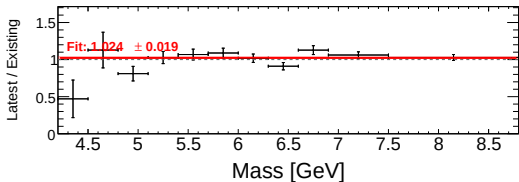
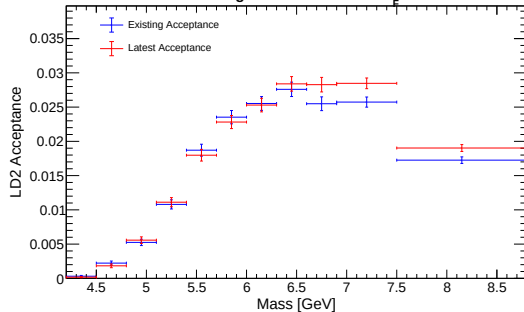


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.10 \leq x_F < 0.15$)

LH2 Existing vs Latest: Mass in x_F Bin 2

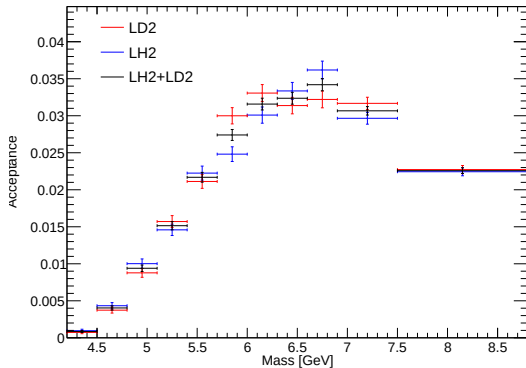


LD2 Existing vs Latest: Mass in x_F Bin 2

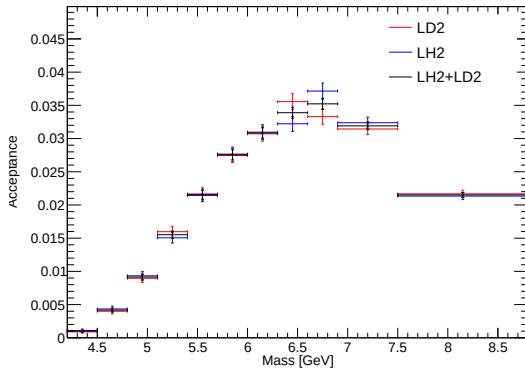


Existing vs Latest: Mass Overlay ($0.15 \leq x_F < 0.20$)

$0.15 \leq x_F < 0.20$

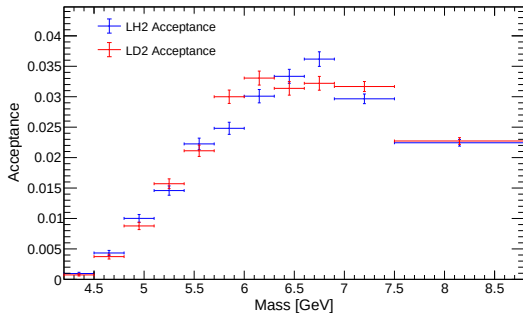


$0.15 \leq x_F < 0.20$

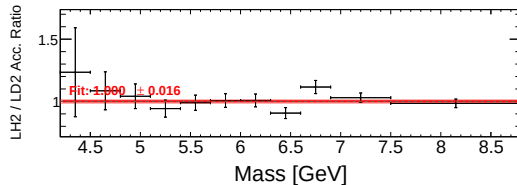
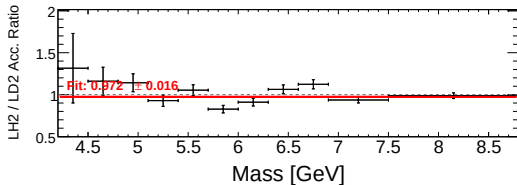
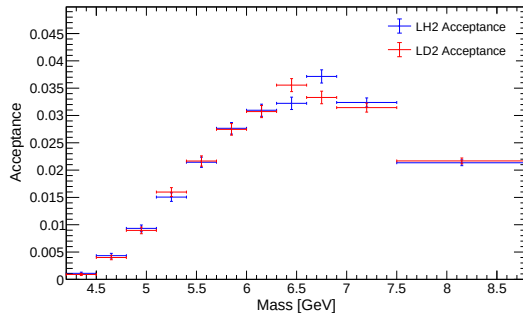


Existing vs Latest: Mass Acceptance ($0.15 \leq x_F < 0.20$)

Invariant Mass Acceptance ($0.15 \leq x_F < 0.20$)

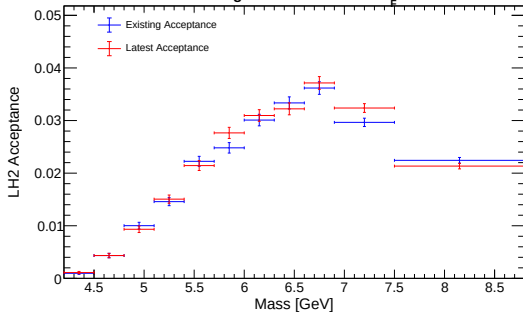


Invariant Mass Acceptance ($0.15 \leq x_F < 0.20$)

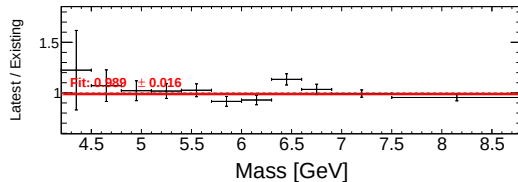
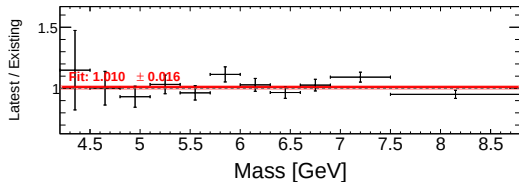
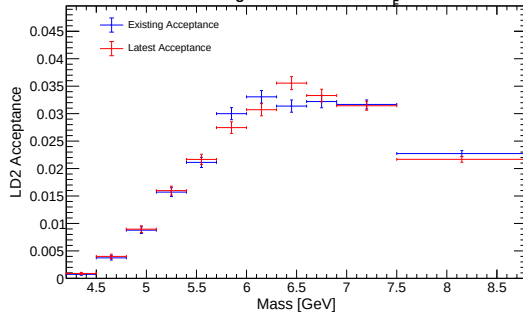


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.15 \leq x_F < 0.20$)

LH2 Existing vs Latest: Mass in x_F Bin 3

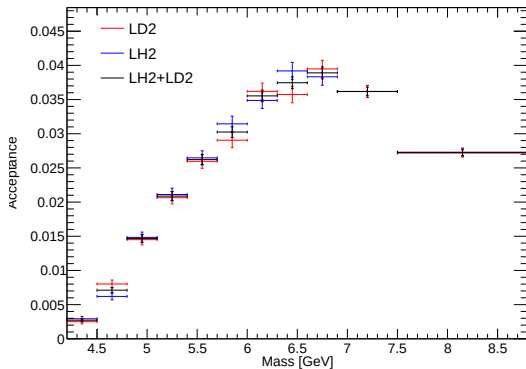


LD2 Existing vs Latest: Mass in x_F Bin 3

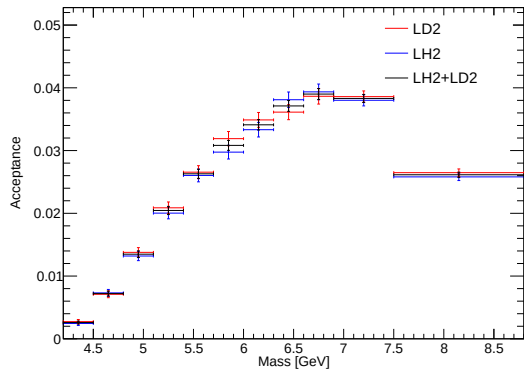


Existing vs Latest: Mass Overlay ($0.20 \leq x_F < 0.25$)

$0.20 \leq x_F < 0.25$

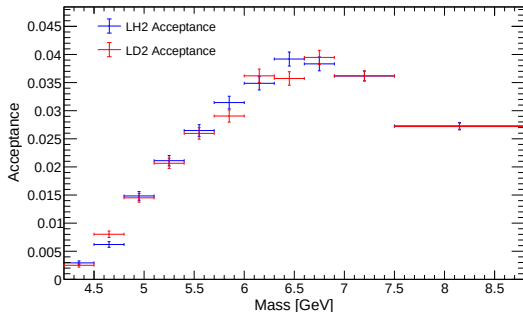


$0.20 \leq x_F < 0.25$

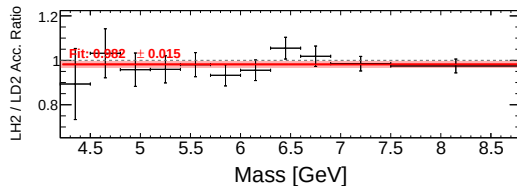
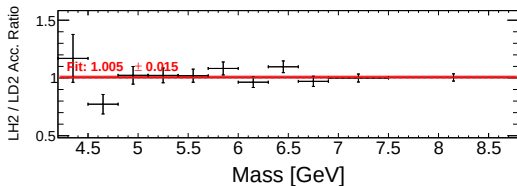
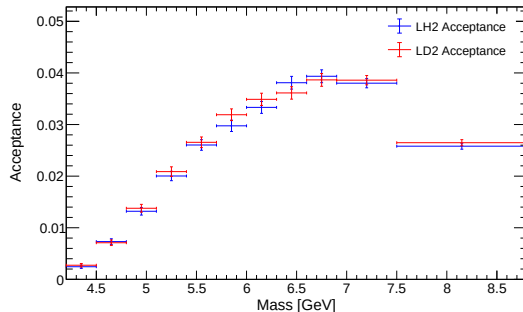


Existing vs Latest: Mass Acceptance ($0.20 \leq x_F < 0.25$)

Invariant Mass Acceptance ($0.20 \leq x_F < 0.25$)

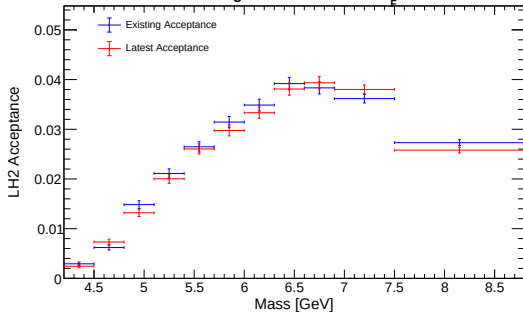


Invariant Mass Acceptance ($0.20 \leq x_F < 0.25$)

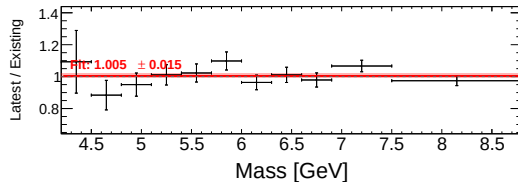
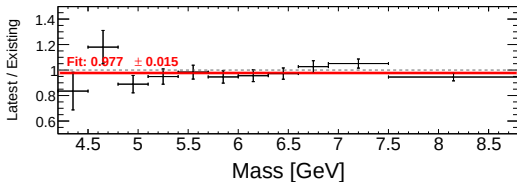
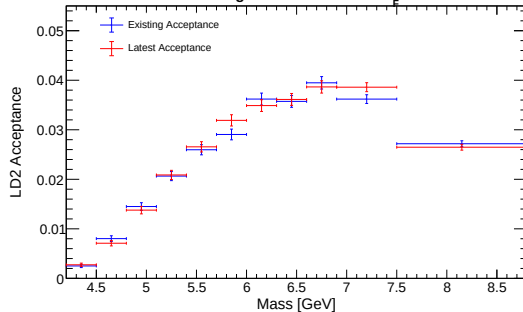


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.20 \leq x_F < 0.25$)

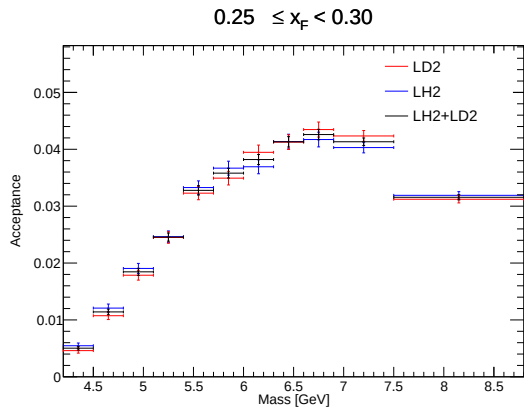
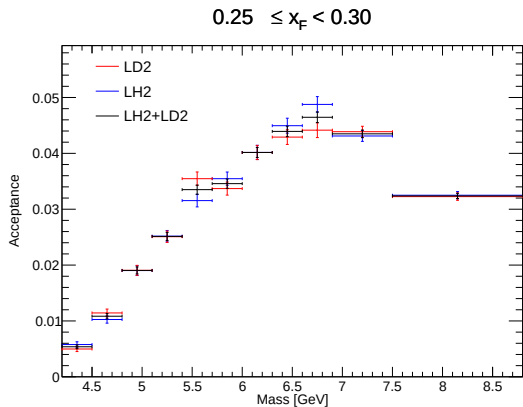
LH2 Existing vs Latest: Mass in x_F Bin 4



LD2 Existing vs Latest: Mass in x_F Bin 4

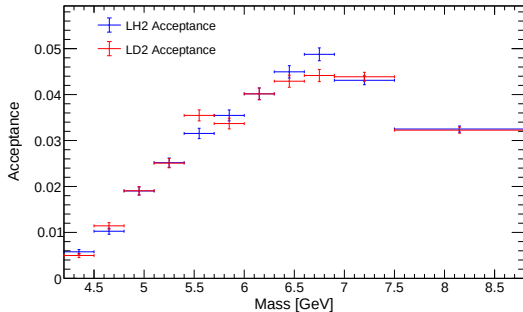


Existing vs Latest: Mass Overlay ($0.25 \leq x_F < 0.30$)

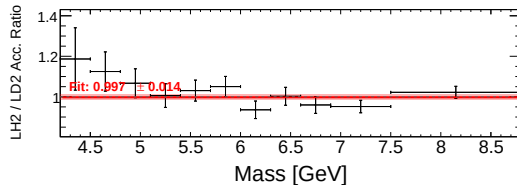
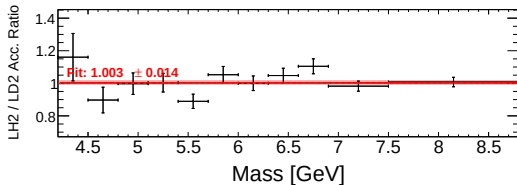
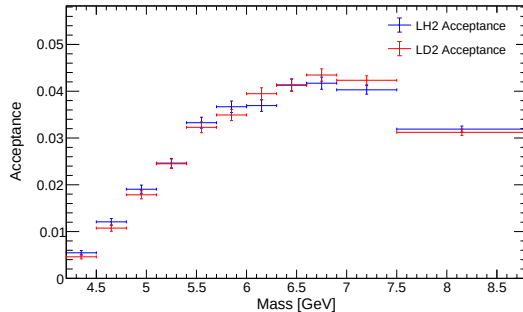


Existing vs Latest: Mass Acceptance ($0.25 \leq x_F < 0.30$)

Invariant Mass Acceptance ($0.25 \leq x_F < 0.30$)

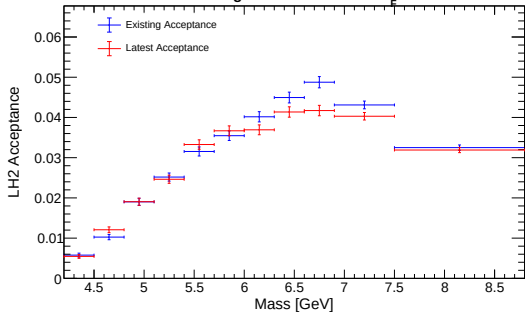


Invariant Mass Acceptance ($0.25 \leq x_F < 0.30$)

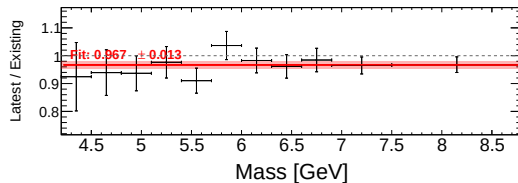
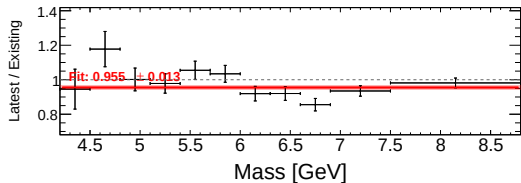
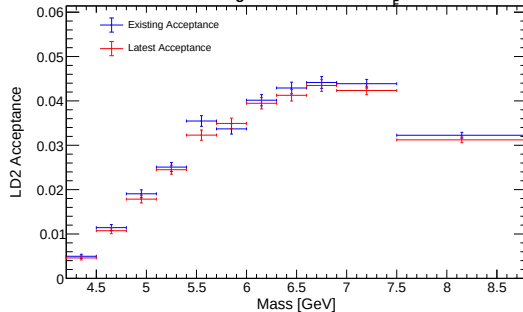


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.25 \leq x_F < 0.30$)

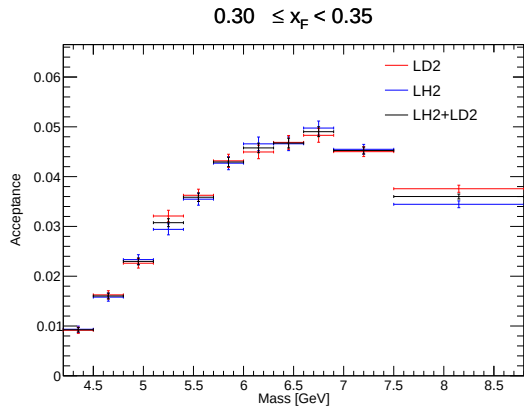
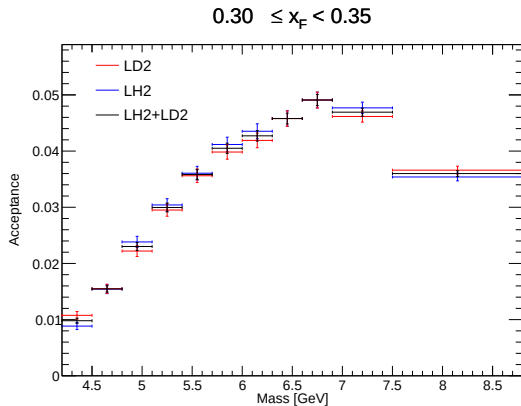
LH2 Existing vs Latest: Mass in x_F Bin 5



LD2 Existing vs Latest: Mass in x_F Bin 5

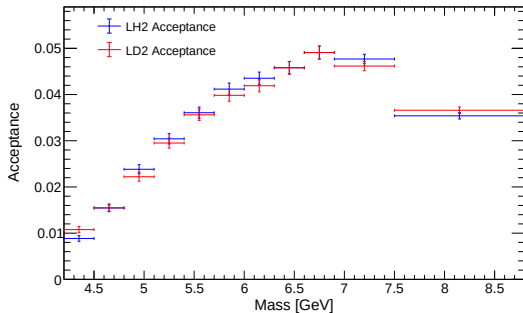


Existing vs Latest: Mass Overlay ($0.30 \leq x_F < 0.35$)

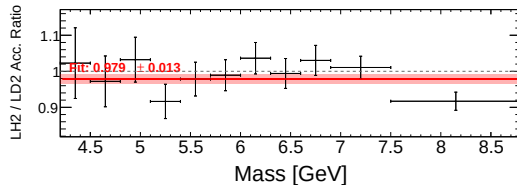
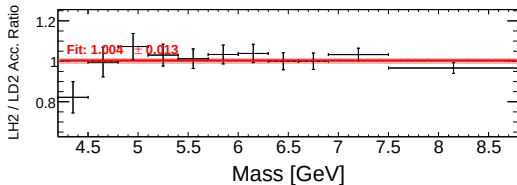
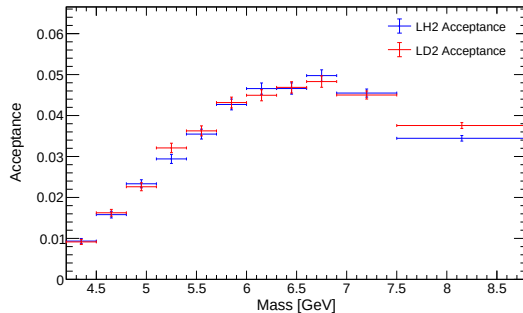


Existing vs Latest: Mass Acceptance ($0.30 \leq x_F < 0.35$)

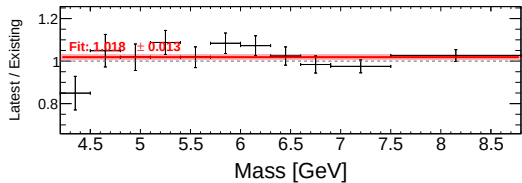
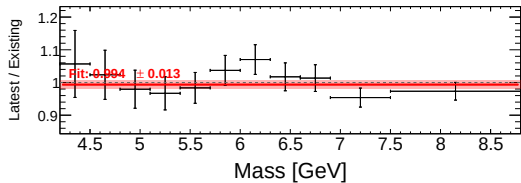
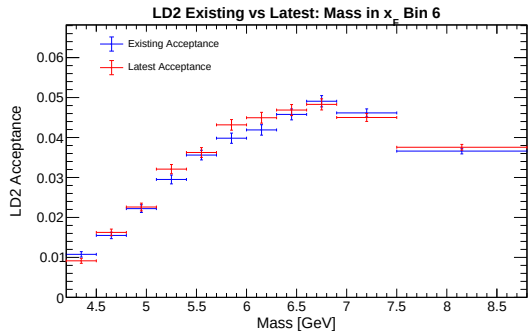
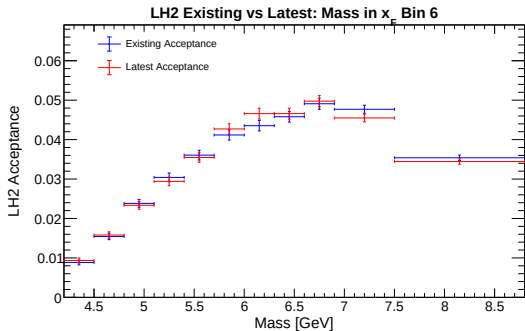
Invariant Mass Acceptance ($0.30 \leq x_F < 0.35$)



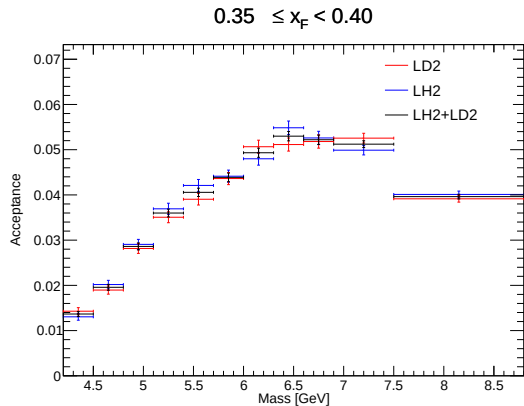
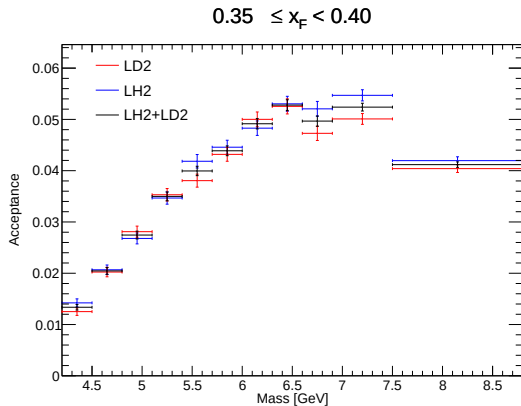
Invariant Mass Acceptance ($0.30 \leq x_F < 0.35$)



LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.30 \leq x_F < 0.35$)

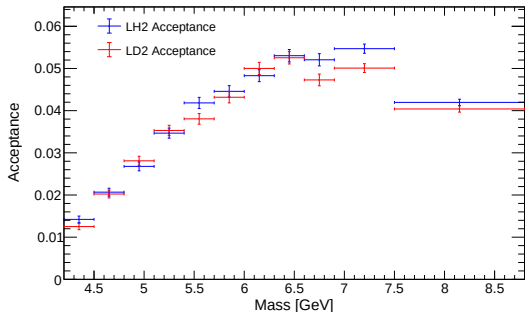


Existing vs Latest: Mass Overlay ($0.35 \leq x_F < 0.40$)

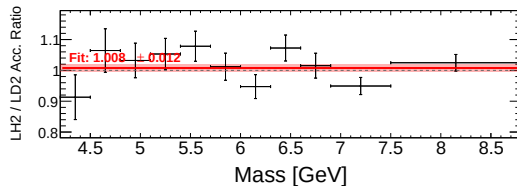
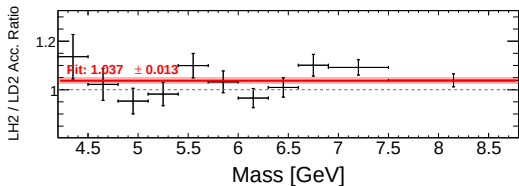
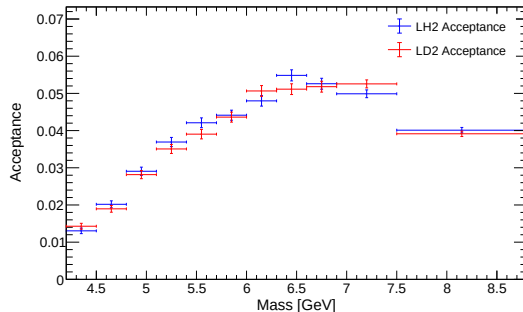


Existing vs Latest: Mass Acceptance ($0.35 \leq x_F < 0.40$)

Invariant Mass Acceptance ($0.35 \leq x_F < 0.40$)

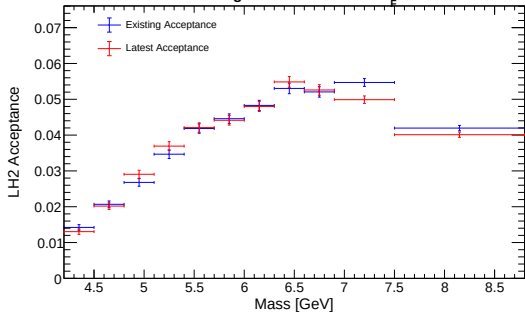


Invariant Mass Acceptance ($0.35 \leq x_F < 0.40$)

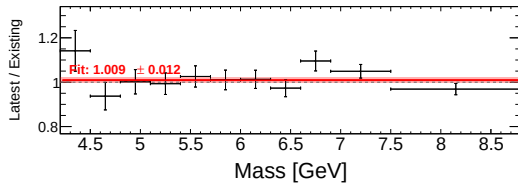
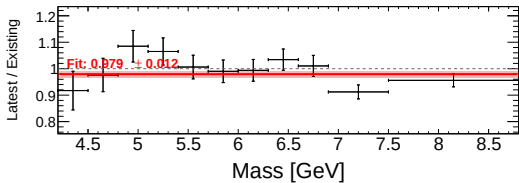
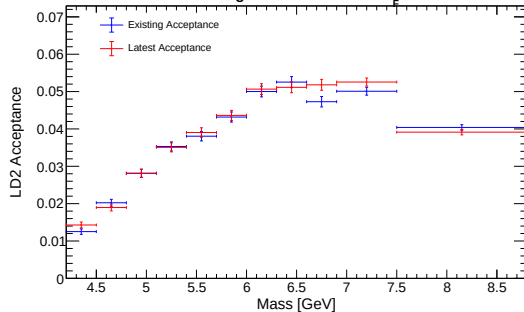


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.35 \leq x_F < 0.40$)

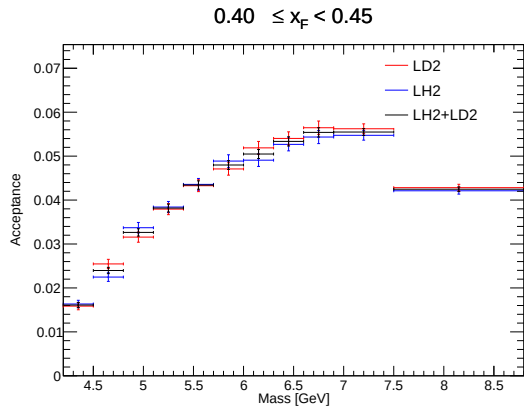
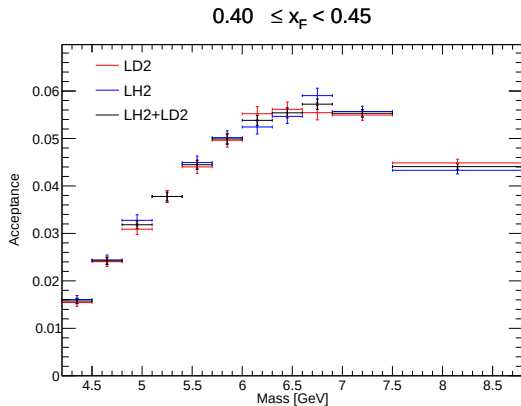
LH2 Existing vs Latest: Mass in x_F Bin 7



LD2 Existing vs Latest: Mass in x_F Bin 7

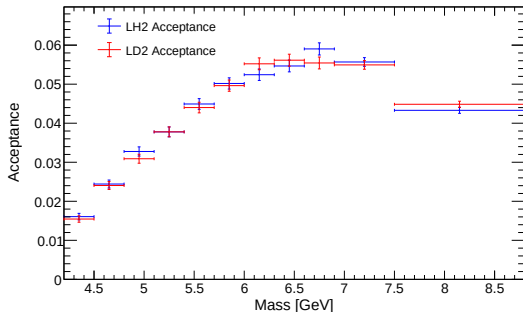


Existing vs Latest: Mass Overlay ($0.40 \leq x_F < 0.45$)

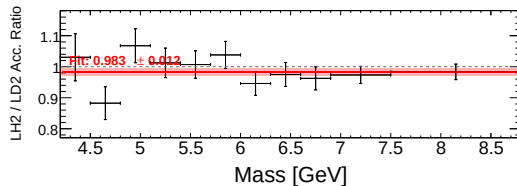
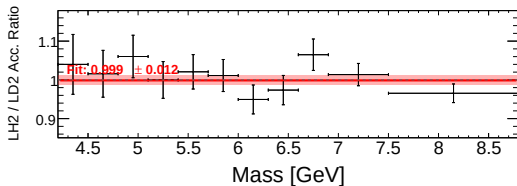
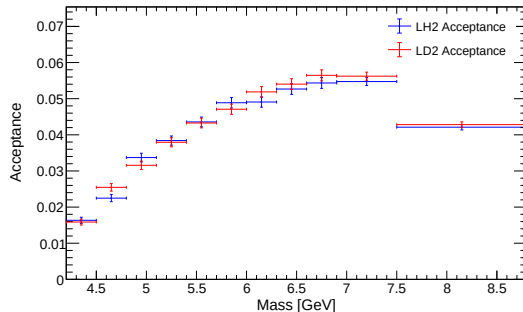


Existing vs Latest: Mass Acceptance ($0.40 \leq x_F < 0.45$)

Invariant Mass Acceptance ($0.40 \leq x_F < 0.45$)

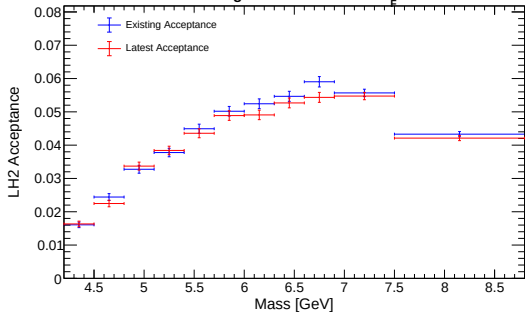


Invariant Mass Acceptance ($0.40 \leq x_F < 0.45$)

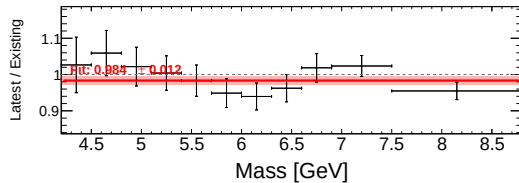
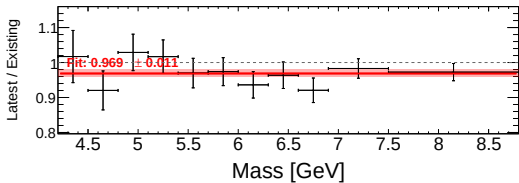
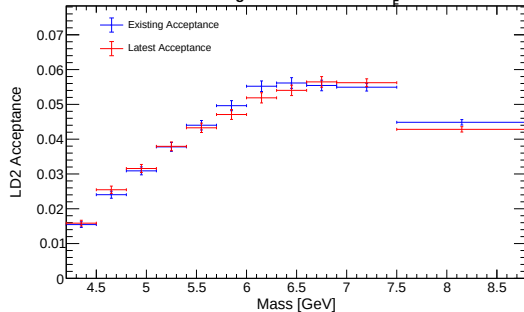


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.40 \leq x_F < 0.45$)

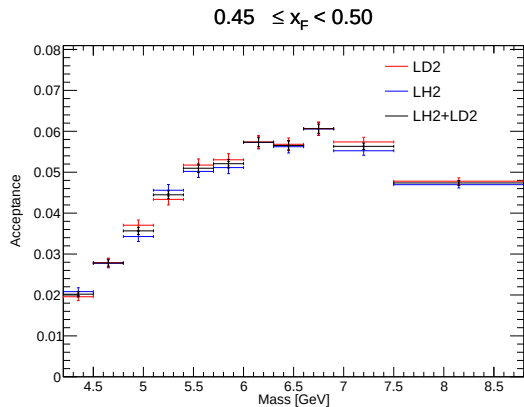
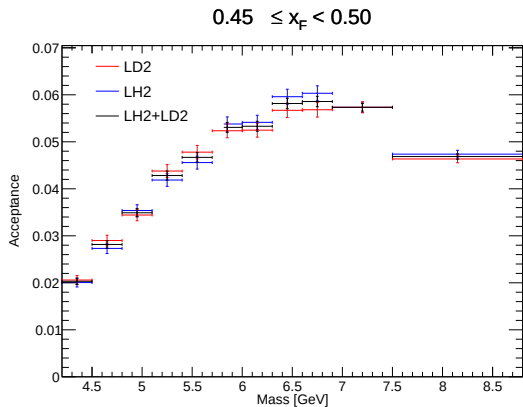
LH2 Existing vs Latest: Mass in x_F Bin 8



LD2 Existing vs Latest: Mass in x_F Bin 8

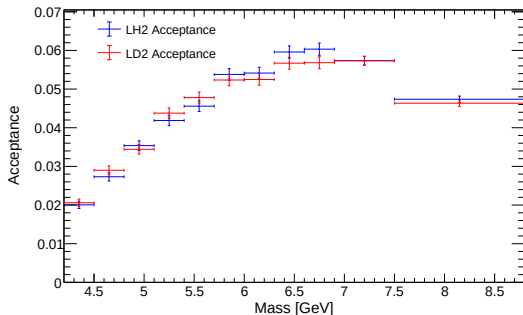


Existing vs Latest: Mass Overlay ($0.45 \leq x_F < 0.50$)

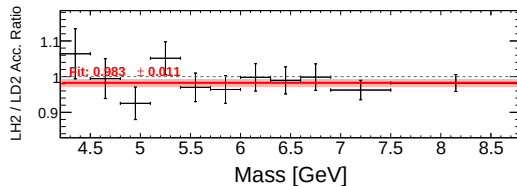
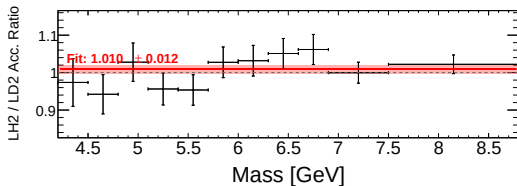
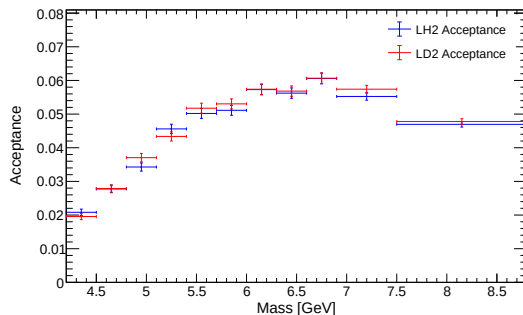


Existing vs Latest: Mass Acceptance ($0.45 \leq x_F < 0.50$)

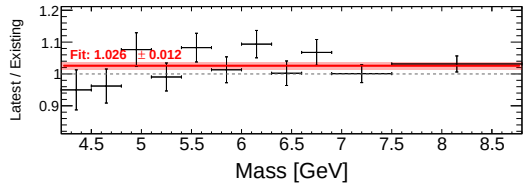
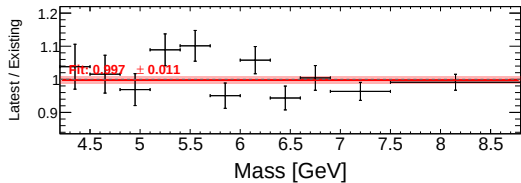
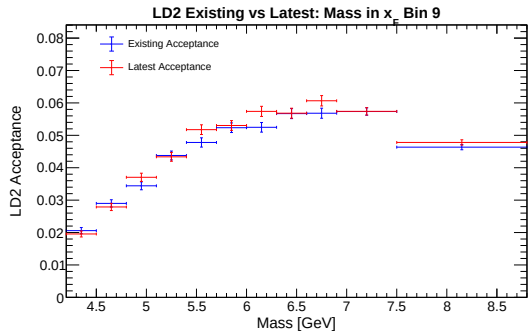
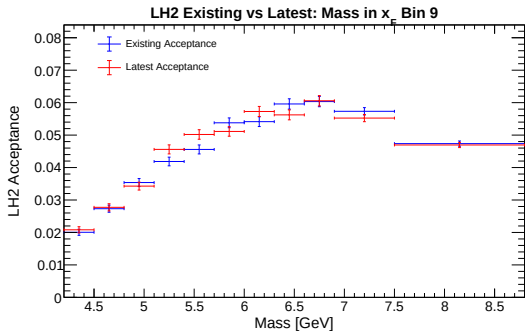
Invariant Mass Acceptance ($0.45 \leq x_F < 0.50$)



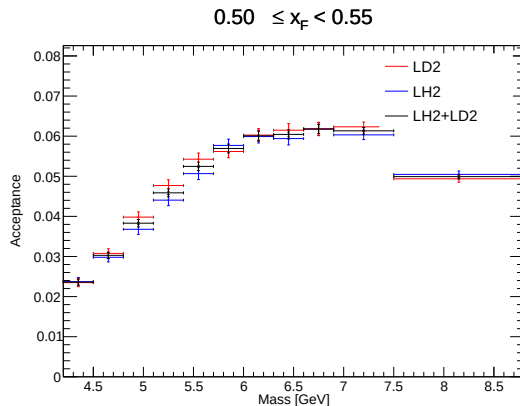
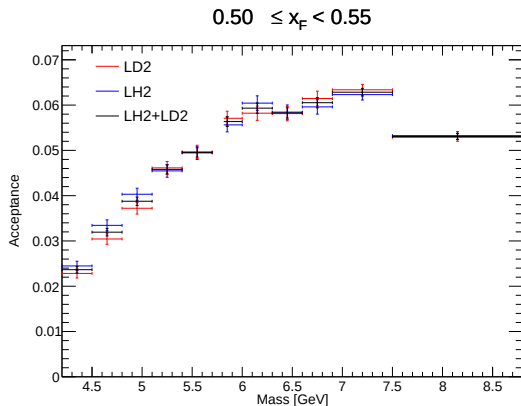
Invariant Mass Acceptance ($0.45 \leq x_F < 0.50$)



LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.45 \leq x_F < 0.50$)

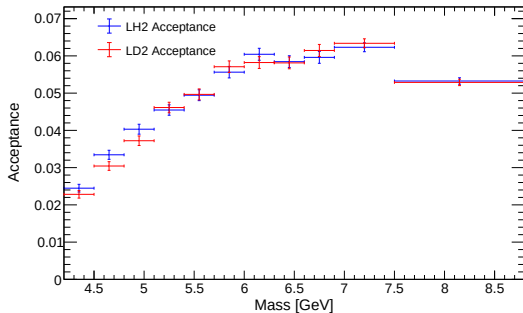


Existing vs Latest: Mass Overlay ($0.50 \leq x_F < 0.55$)

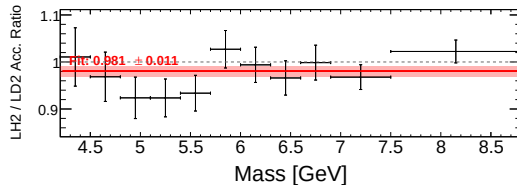
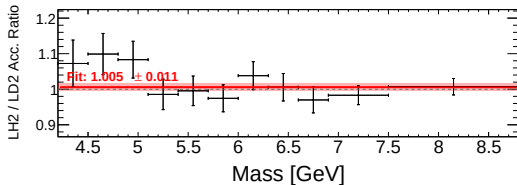
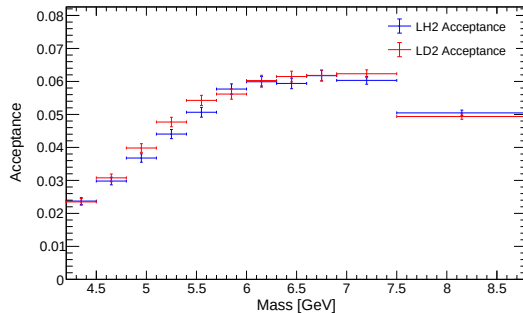


Existing vs Latest: Mass Acceptance ($0.50 \leq x_F < 0.55$)

Invariant Mass Acceptance ($0.50 \leq x_F < 0.55$)

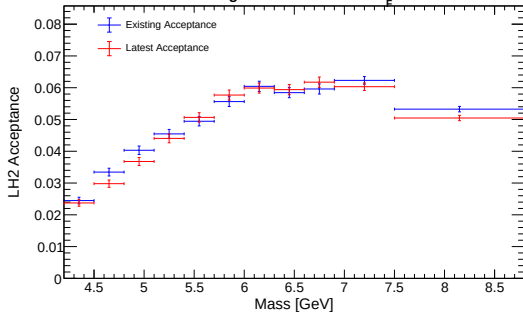


Invariant Mass Acceptance ($0.50 \leq x_F < 0.55$)

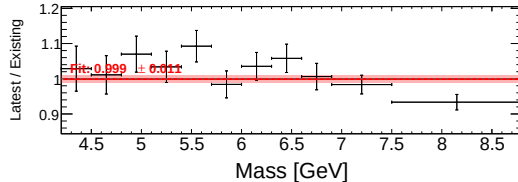
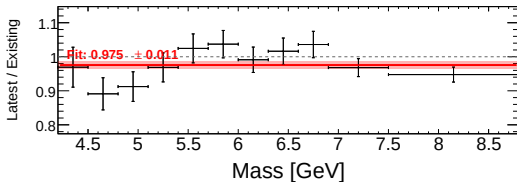
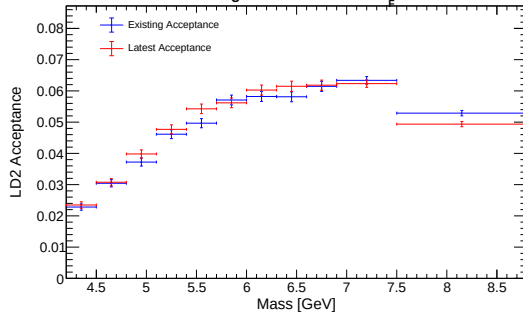


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.50 \leq x_F < 0.55$)

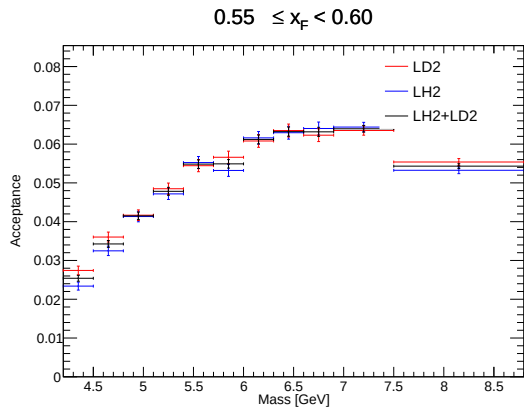
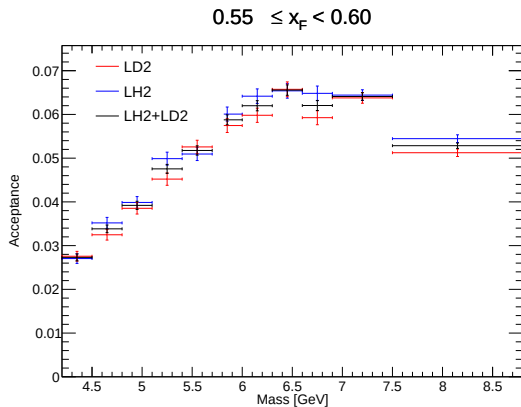
LH2 Existing vs Latest: Mass in x_F Bin 10



LD2 Existing vs Latest: Mass in x_F Bin 10

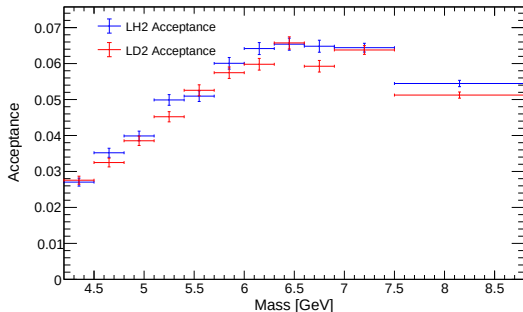


Existing vs Latest: Mass Overlay ($0.55 \leq x_F < 0.60$)

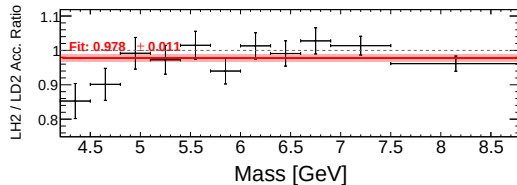
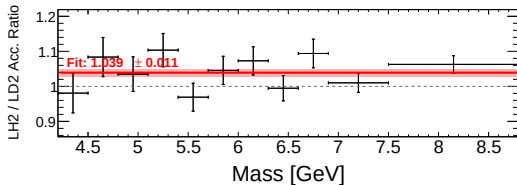
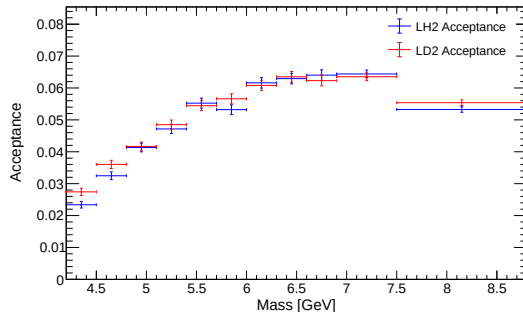


Existing vs Latest: Mass Acceptance ($0.55 \leq x_F < 0.60$)

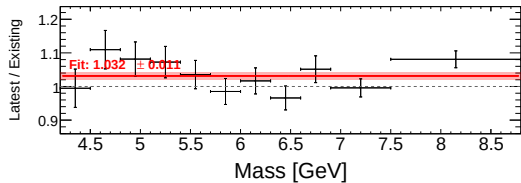
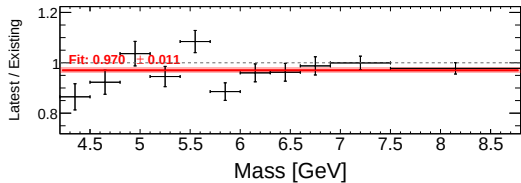
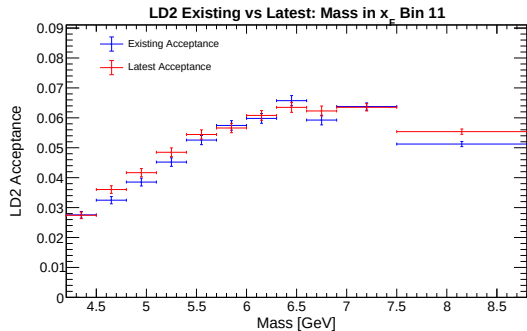
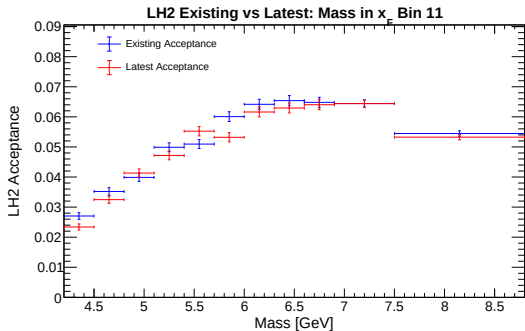
Invariant Mass Acceptance ($0.55 \leq x_F < 0.60$)



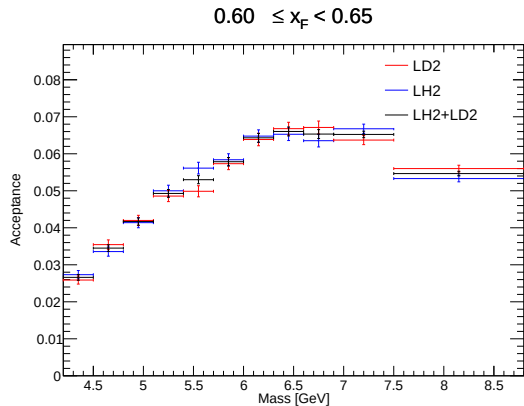
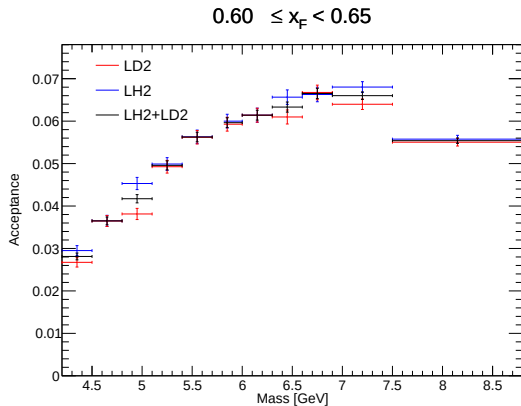
Invariant Mass Acceptance ($0.55 \leq x_F < 0.60$)



LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.55 \leq x_F < 0.60$)

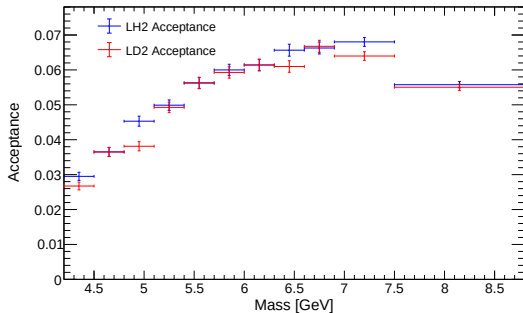


Existing vs Latest: Mass Overlay ($0.60 \leq x_F < 0.65$)

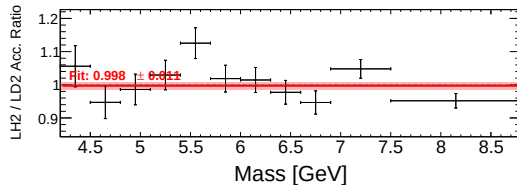
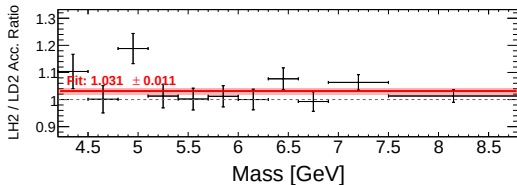
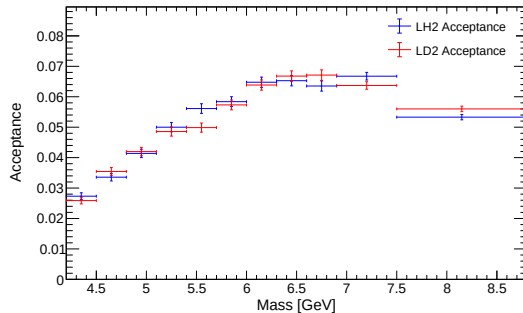


Existing vs Latest: Mass Acceptance ($0.60 \leq x_F < 0.65$)

Invariant Mass Acceptance ($0.60 \leq x_F < 0.65$)

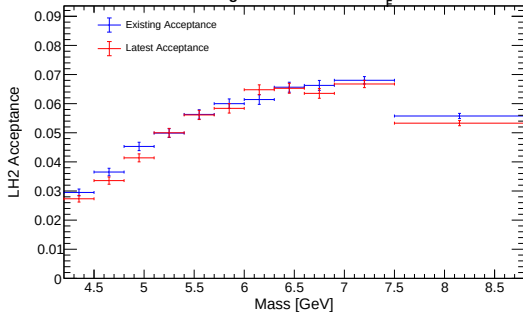


Invariant Mass Acceptance ($0.60 \leq x_F < 0.65$)

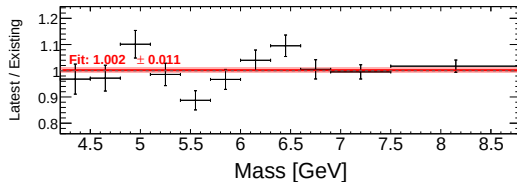
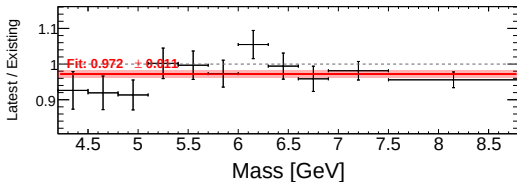
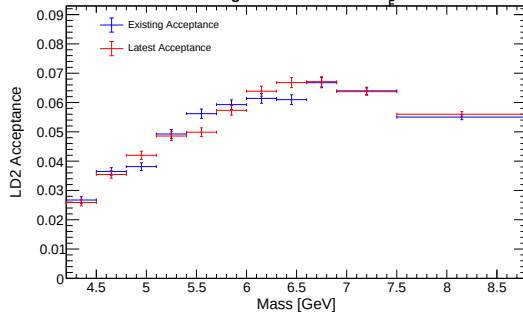


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.60 \leq x_F < 0.65$)

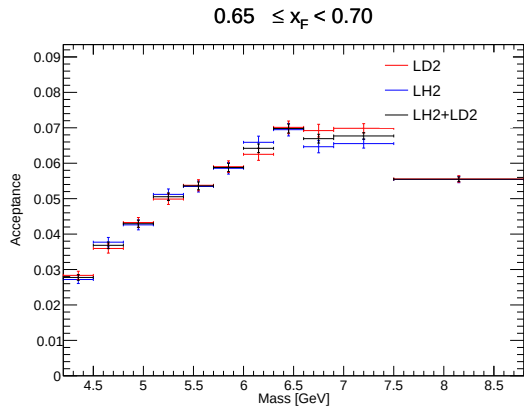
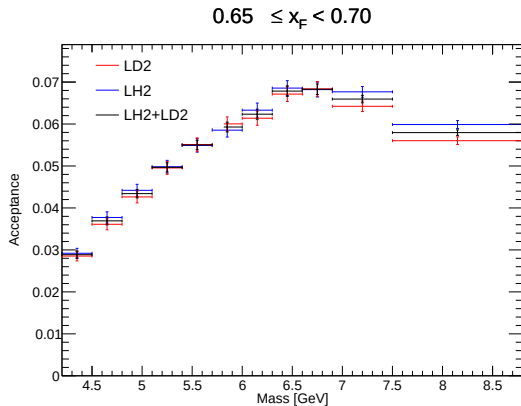
LH2 Existing vs Latest: Mass in x_F Bin 12



LD2 Existing vs Latest: Mass in x_F Bin 12

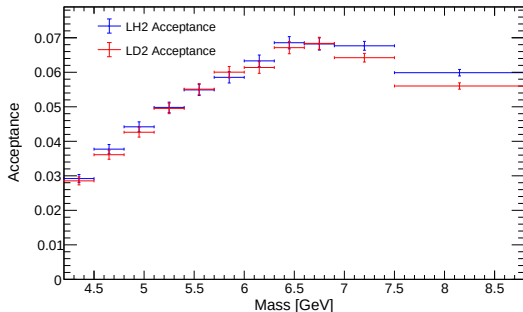


Existing vs Latest: Mass Overlay ($0.65 \leq x_F < 0.70$)

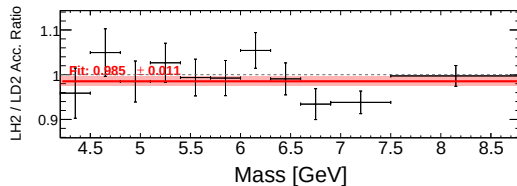
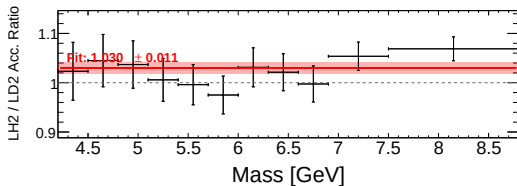
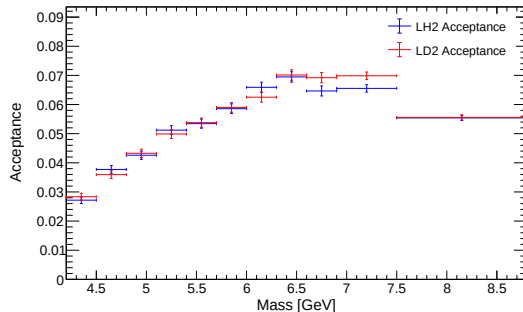


Existing vs Latest: Mass Acceptance ($0.65 \leq x_F < 0.70$)

Invariant Mass Acceptance ($0.65 \leq x_F < 0.70$)

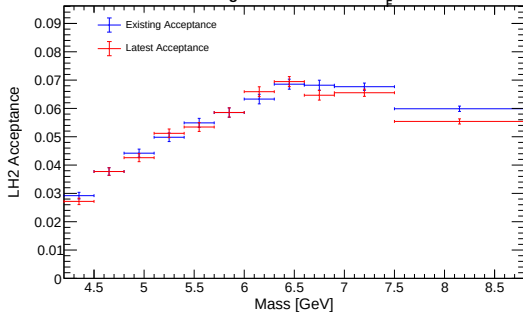


Invariant Mass Acceptance ($0.65 \leq x_F < 0.70$)

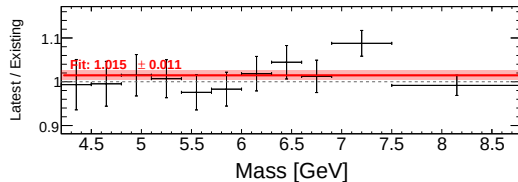
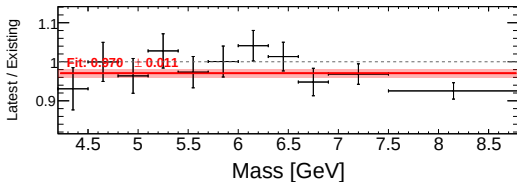
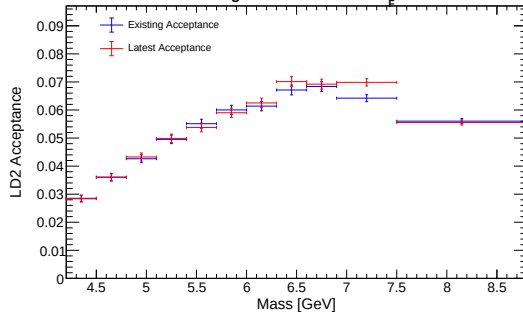


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.65 \leq x_F < 0.70$)

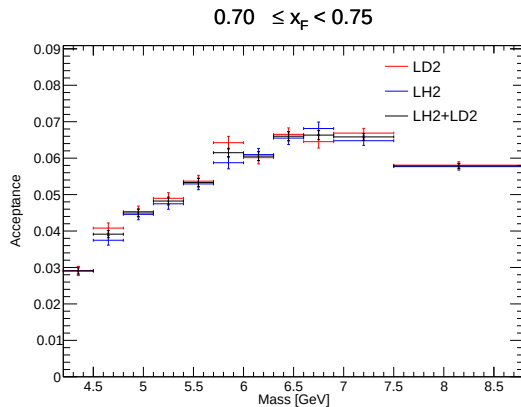
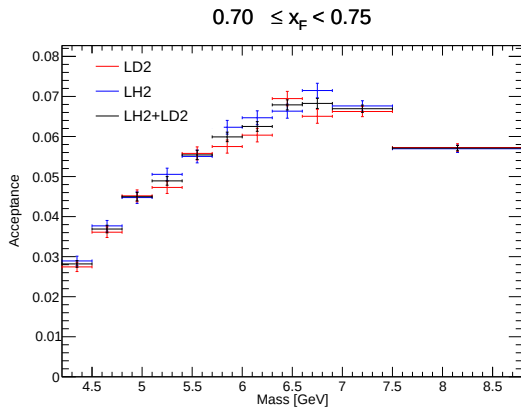
LH2 Existing vs Latest: Mass in x_F Bin 13



LD2 Existing vs Latest: Mass in x_F Bin 13

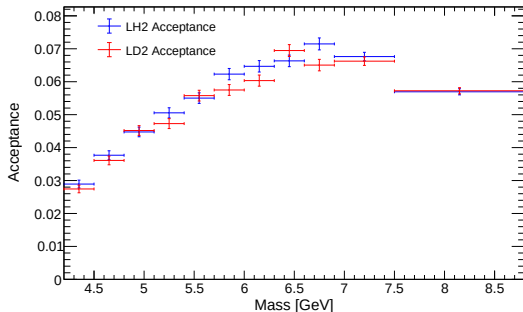


Existing vs Latest: Mass Overlay ($0.70 \leq x_F < 0.75$)

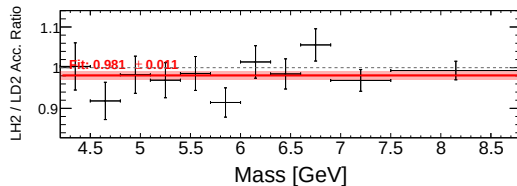
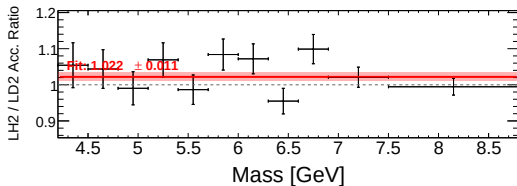
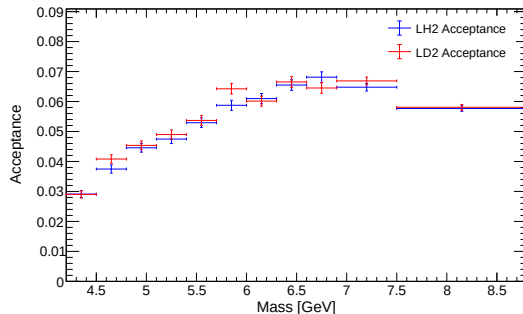


Existing vs Latest: Mass Acceptance ($0.70 \leq x_F < 0.75$)

Invariant Mass Acceptance ($0.70 \leq x_F < 0.75$)

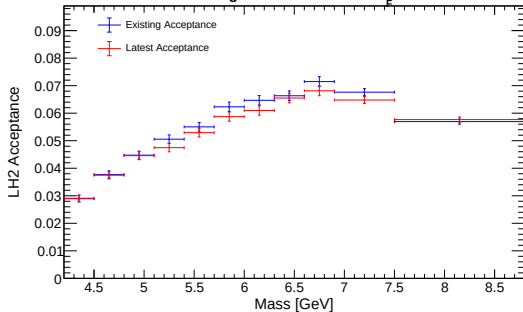


Invariant Mass Acceptance ($0.70 \leq x_F < 0.75$)

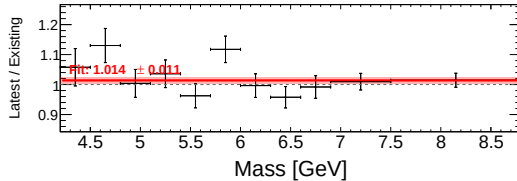
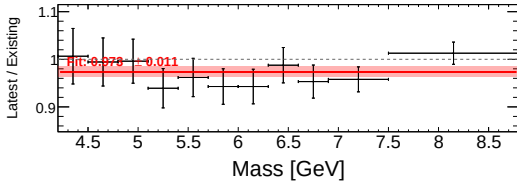
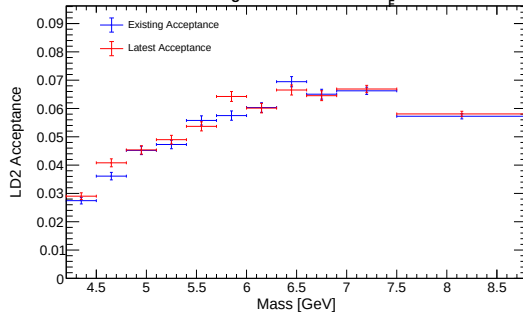


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.70 \leq x_F < 0.75$)

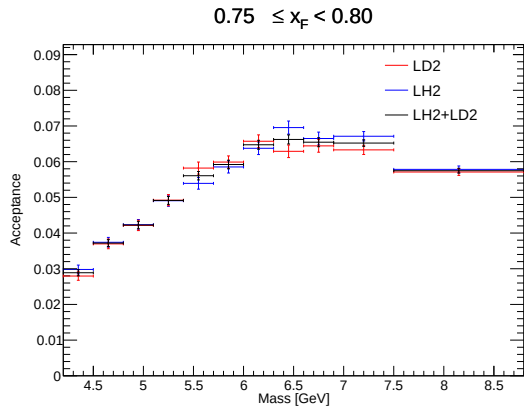
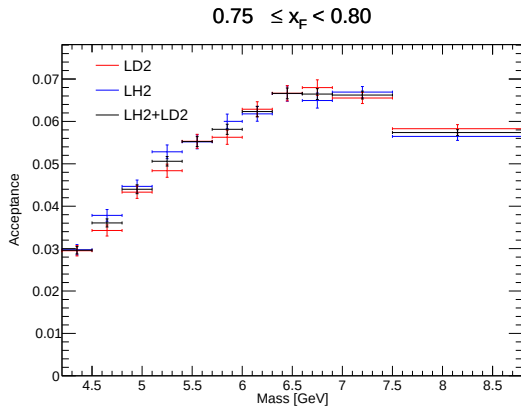
LH2 Existing vs Latest: Mass in x_F Bin 14



LD2 Existing vs Latest: Mass in x_F Bin 14

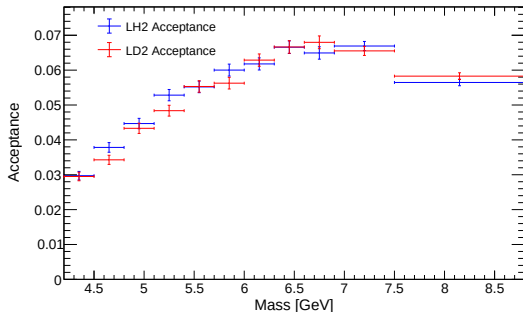


Existing vs Latest: Mass Overlay ($0.75 \leq x_F < 0.80$)

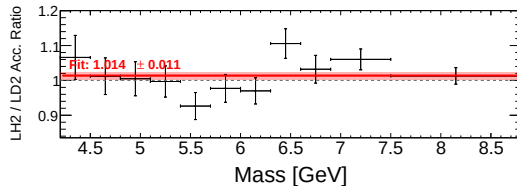
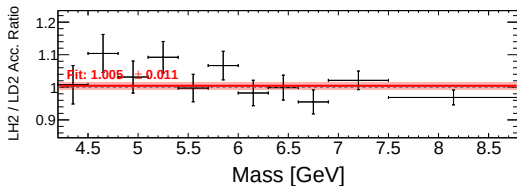
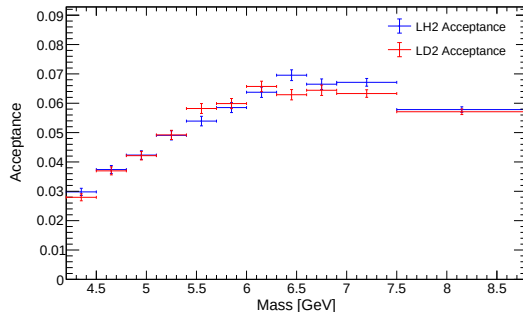


Existing vs Latest: Mass Acceptance ($0.75 \leq x_F < 0.80$)

Invariant Mass Acceptance ($0.75 \leq x_F < 0.80$)

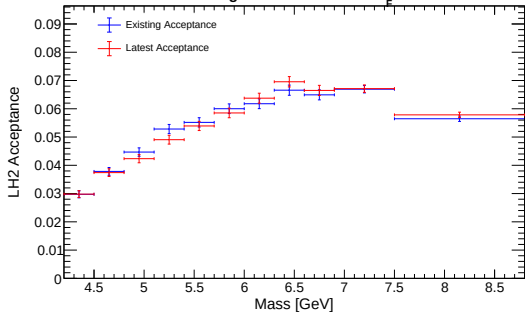


Invariant Mass Acceptance ($0.75 \leq x_F < 0.80$)

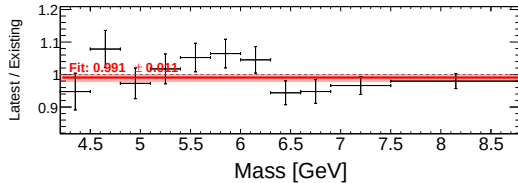
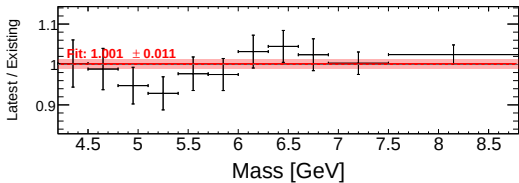
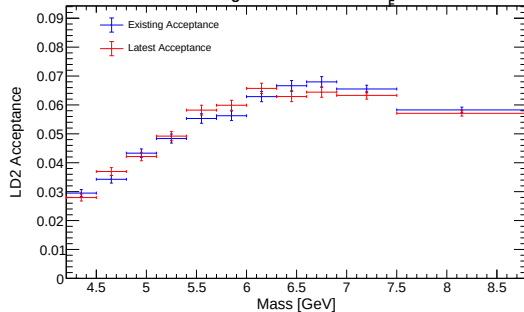


LH2 (L) vs LD2 (R): Latest/Existing Mass Acc. ($0.75 \leq x_F < 0.80$)

LH2 Existing vs Latest: Mass in x_F Bin 15



LD2 Existing vs Latest: Mass in x_F Bin 15



- Acceptance corrections calculated for the double differential cross-section study was minimally impacted.
- Acceptance Vs x_F and Acceptance Vs p_T plots demonstrate larger differences.